

Performance Testing

Date	27 June 2026
Team ID	SWUID2026217448
Project Name	A Comprehensive Measure of Well-Being - HDI Predictor
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Manual Testing / Browser (Chrome)
Type of Testing	Functional Testing
Target Module / API	/predict endpoint - HDI Prediction Form
Test Environment	Local - Flask Development Server (localhost:5000)
Test Date	27 June 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Predict HDI for high income country (Life Exp=80, EYS=16, MYS=13, GNI=60000)	1	2	Very High category predicted correctly
2	Predict HDI for low income country (Life Exp=55, EYS=8, MYS=3, GNI=1500)	1	2	Low category predicted correctly
3	Predict HDI for medium country (Life Exp=70, EYS=12, MYS=8, GNI=8000)	1	2	Medium category predicted correctly
4	Test India-like values (Life Exp=70, EYS=12, MYS=6.5, GNI=6590)	1	2	Medium/High category predicted

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	< 1 second	Pass	Flask responds instantly
2	Response Time (Max)	< 5 seconds	< 2 seconds	Pass	Well within limit
3	Throughput (Req/sec)	N/A	1 req/sec	Pass	Single user local app
4	Error Rate	< 1%	0%	Pass	No errors encountered
5	CPU Utilization	< 80%	~15%	Pass	Low CPU usage
6	Memory Utilization	< 80%	~200MB	Pass	Low memory usage

Step 4: Observations & Analysis

Key Findings:

Flask app responds in under 1 second.

Bottlenecks Identified:

None identified.

Optimization Steps Taken:

No optimization needed - performance is excellent.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

Human Development Index Predictor

Enter Country Details:

Life Expectancy:

Expected Years of Schooling:

Mean Years of Schooling:

GNI Per Capita:

Predict HDI

Human Development Index Predictor

Enter Country Details:

Life Expectancy:

75.5

Expected Years of Schooling:

14.2

Mean Years of Schooling:

10.5

GNI Per Capita:

15000

Predict HDI

HDI Prediction Result

0.78

Category: High

Predict Another